ADI-FDTD for Inhomogeneous Media

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1 Preliminaries

1.1 Maxwell's Equations

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} \tag{1.1a}$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \tag{1.1b}$$

Expanding (1.1a) into components:

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \varepsilon \frac{\partial E_x}{\partial t} + \sigma E_x \tag{1.2a}$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = \varepsilon \frac{\partial E_y}{\partial t} + \sigma E_y \tag{1.2b}$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \varepsilon \frac{\partial E_z}{\partial t} + \sigma E_z \tag{1.2c}$$

Expanding (1.1b) into components:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu}\left(\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y}\right) \tag{1.3a}$$

$$\frac{\partial H_y}{\partial t} = \frac{1}{\mu}\left(\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z}\right) \tag{1.3b}$$

$$\frac{\partial H_z}{\partial t} = \frac{1}{\mu}\left(\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x}\right) \tag{1.3c}$$

1.2 Yee Grid in AMReX

The Yee grid stores different field components at different physical locations within each computational cell. In AMReX this staggering is represented by the `IndexType` used to define each `MultiFab`: a cell-centered direction has index type 0 and is located at $i + \frac{1}{2}$, while a nodal direction has index type 1 and is located at i. The implementation uses

$$\mathbf{E}_\alpha : \text{cell-centered in the } \alpha \text{ direction and nodal in the other two directions,}$$
$$\mathbf{H}_\alpha : \text{nodal in the } \alpha \text{ direction and cell-centered in the other two directions.}$$

Thus the component locations are

Component	x location	y location	z location
$E_{x,i,j,k}$	$i + \frac{1}{2}$	j	k
$E_{y,i,j,k}$	i	$j + \frac{1}{2}$	k
$E_{z,i,j,k}$	i	j	$k + \frac{1}{2}$
$H_{x,i,j,k}$	i	$j + \frac{1}{2}$	$k + \frac{1}{2}$
$H_{y,i,j,k}$	$i + \frac{1}{2}$	j	$k + \frac{1}{2}$
$H_{z,i,j,k}$	$i + \frac{1}{2}$	$j + \frac{1}{2}$	k

The half-cell offsets therefore do not appear explicitly as array indices such as $i+\frac{1}{2}$. They are encoded in the `MultiFab` staggering. The integer offsets in the finite-difference stencil choose the two source values that bracket the field being updated. For example, $H_{x,i,j,k}$ is located at $j+\frac{1}{2}$ in the y direction, while $E_{z,i,j,k}$ and $E_{z,i,j+1,k}$ are located at j and $j+1$. Therefore

$$\left.\frac{\partial E_z}{\partial y}\right|_{H_{x,i,j,k}} \approx \frac{E_{z,i,j+1,k} - E_{z,i,j,k}}{\Delta y}. \tag{1.4}$$

Similarly, $E_{x,i,j,k}$ is located at j, while $H_{z,i,j-1,k}$ and $H_{z,i,j,k}$ are located at $j-\frac{1}{2}$ and $j+\frac{1}{2}$, so

$$\left.\frac{\partial H_z}{\partial y}\right|_{E_{x,i,j,k}} \approx \frac{H_{z,i,j,k} - H_{z,i,j-1,k}}{\Delta y}. \tag{1.5}$$

This is why the magnetic update below uses forward differences such as $j+1$ and $k+1$, while the electric update uses backward differences such as $j-1$ and $k-1$: each derivative is centered at the staggered location of the field component being updated.

2 FDTD Method

The explicit Yee FDTD update used here is written in three stages, matching the implementation: first update $\mathbf{H}$ by a half step, then update $\mathbf{E}$ by a full step, and finally update $\mathbf{H}$ by another half step. Define the update coefficients

$$C_a = \frac{2\varepsilon - \sigma\Delta t}{2\varepsilon + \sigma\Delta t}, \qquad C_b = \frac{2\Delta t}{2\varepsilon + \sigma\Delta t}, \qquad D_b = \frac{\Delta t}{\mu}, \tag{2.1}$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity.

2.1 First Magnetic Half-Step

$$H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - \frac{D^{n+\frac{1}{2}}_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{2}\left[\frac{E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta y} - \frac{E^{n}_{y,i,j+\frac{1}{2},k+1} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta z}\right] \tag{2.2a}$$

$$H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - \frac{D^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}}}{2}\left[\frac{E^{n}_{x,i+\frac{1}{2},j,k+1} - E^{n}_{x,i+\frac{1}{2},j,k}}{\Delta z} - \frac{E^{n}_{z,i+1,j,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta x}\right] \tag{2.2b}$$

$$H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - \frac{D^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k}}{2}\left[\frac{E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta x} - \frac{E^{n}_{x,i+\frac{1}{2},j+1,k} - E^{n}_{x,i+\frac{1}{2},j,k}}{\Delta y}\right]. \tag{2.2c}$$

In the AMReX Yee grid, with homogeneous media, the magnetic field updates are implemented as

$$B_x(i,j,k) - = \frac{1}{2\Delta t}\left[\frac{E_z(i,j+1,k) - E_z(i,j,k)}{\Delta y} - \frac{E_y(i,j,k+1) - E_y(i,j,k)}{\Delta z}\right] \tag{2.3a}$$

$$B_y(i,j,k) - = \frac{1}{2\Delta t}\left[\frac{E_x(i,j,k+1) - E_x(i,j,k)}{\Delta z} - \frac{E_z(i+1,j,k) - E_z(i,j,k)}{\Delta x}\right] \tag{2.3b}$$

$$B_z(i,j,k) - = \frac{1}{2\Delta t}\left[\frac{E_y(i+1,j,k) - E_y(i,j,k)}{\Delta x} - \frac{E_x(i,j+1,k) - E_x(i,j,k)}{\Delta y}\right]. \tag{2.3c}$$

2.2 Electric Full-Step

$$E^{n+1}_{x,i+\frac{1}{2},j,k} = C^{n+1}_{a,i+\frac{1}{2},j,k}E^{n}_{x,i+\frac{1}{2},j,k} + C^{n+1}_{b,i+\frac{1}{2},j,k}\left[\frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}\right] \tag{2.4a}$$

$$E^{n+1}_{y,i,j+\frac{1}{2},k} = C^{n+1}_{a,i,j+\frac{1}{2},k}E^{n}_{y,i,j+\frac{1}{2},k} + C^{n+1}_{b,i,j+\frac{1}{2},k}\left[\frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}\right] \tag{2.4b}$$

$$E^{n+1}_{z,i,j,k+\frac{1}{2}} = C^{n+1}_{a,i,j,k+\frac{1}{2}}E^{n}_{z,i,j,k+\frac{1}{2}} + C^{n+1}_{b,i,j,k+\frac{1}{2}}\left[\frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}\right]. \tag{2.4c}$$

In the AMReX Yee grid, with homogeneous media, the electric field updates are implemented as

$$E_x(i,j,k) += c^2\Delta t\left[\frac{B_z(i,j,k)-B_z(i,j-1,k)}{\Delta y}-\frac{B_y(i,j,k)-B_y(i,j,k-1)}{\Delta z}\right] \tag{2.5a}$$

$$E_y(i,j,k) += c^2\Delta t\left[\frac{B_x(i,j,k)-B_x(i,j,k-1)}{\Delta z}-\frac{B_z(i,j,k)-B_z(i-1,j,k)}{\Delta x}\right] \tag{2.5b}$$

$$E_z(i,j,k) += c^2\Delta t\left[\frac{B_y(i,j,k)-B_y(i-1,j,k)}{\Delta x}-\frac{B_x(i,j,k)-B_x(i,j-1,k)}{\Delta y}\right]. \tag{2.5c}$$

2.3 Second Magnetic Half-Step

$$H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - \frac{D_b}{2}\left[\frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}}-E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y}-\frac{E^{n+1}_{y,i,j+\frac{1}{2},k+1}-E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta z}\right] \tag{2.6a}$$

$$H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - \frac{D_b}{2}\left[\frac{E^{n+1}_{x,i+\frac{1}{2},j,k+1}-E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta z}-\frac{E^{n+1}_{z,i+1,j,k+\frac{1}{2}}-E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta x}\right] \tag{2.6b}$$

$$H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - \frac{D_b}{2}\left[\frac{E^{n+1}_{y,i+1,j+\frac{1}{2},k}-E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta x}-\frac{E^{n+1}_{x,i+\frac{1}{2},j+1,k}-E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta y}\right]. \tag{2.6c}$$

In the AMReX Yee grid, the second magnetic field updates are implemented as

$$B_x(i,j,k) -= \frac{1}{2\Delta t}\left[\frac{E_z(i,j+1,k)-E_z(i,j,k)}{\Delta y}-\frac{E_y(i,j,k+1)-E_y(i,j,k)}{\Delta z}\right] \tag{2.7a}$$

$$B_y(i,j,k) -= \frac{1}{2\Delta t}\left[\frac{E_x(i,j,k+1)-E_x(i,j,k)}{\Delta z}-\frac{E_z(i+1,j,k)-E_z(i,j,k)}{\Delta x}\right] \tag{2.7b}$$

$$B_z(i,j,k) -= \frac{1}{2\Delta t}\left[\frac{E_y(i+1,j,k)-E_y(i,j,k)}{\Delta x}-\frac{E_x(i,j+1,k)-E_x(i,j,k)}{\Delta y}\right]. \tag{2.7c}$$

This explicit method is simple and local, but the time step is restricted by the CFL stability condition. For a uniform lossless grid, a typical stability limit is

$$\Delta t \le \frac{1}{c\sqrt{\frac{1}{\Delta x^2}+\frac{1}{\Delta y^2}+\frac{1}{\Delta z^2}}}, \qquad c = \frac{1}{\sqrt{\mu\varepsilon}}. \tag{2.8}$$

3 ADI Method

The ADI-FDTD update advances the fields by a full time step Δt in two half steps of size $\Delta t/2$. Unlike the explicit three-stage leapfrog update above, each ADI half step first updates $\mathbf{E}$ implicitly over $\Delta t/2$, then updates $\mathbf{H}$ explicitly over $\Delta t/2$ from the updated electric field. The first half step treats one set of implicit directions; the second half step treats the complementary directions to complete the cycle. Define the half-step update coefficients

$$C_a = \frac{4\varepsilon-\sigma\Delta t}{4\varepsilon+\sigma\Delta t}, \qquad C_b = \frac{2\Delta t}{4\varepsilon+\sigma\Delta t}, \qquad D_b = \frac{\Delta t}{2\mu}, \tag{3.1}$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity. These differ from the full-step coefficients in (2.1) because each electric update uses $\Delta t/2$.

3.1 First ADI Half-Step

E_x and H_z By equation (1.2a),

$$\varepsilon_{i+\frac{1}{2},j,k}\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}-E^{n}_{x,i+\frac{1}{2},j,k}}{\Delta t/2}+\sigma_{i+\frac{1}{2},j,k}\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}+E^{n}_{x,i+\frac{1}{2},j,k}}{2} = [H_1], \tag{3.2}$$

where

$$[H_1] = \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}. \tag{3.3}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_x^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_x^{n} + [H_1]. \tag{3.4}$$

Hence

$$E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k}\,[H_1], \tag{3.5}$$

By equation (1.3c),

$$\begin{aligned} H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k}\Bigg(&\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} \\ &- \frac{E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta x}\Bigg), \end{aligned} \tag{3.6}$$

Substituting the magnetic update into (3.5) and eliminating $H_z^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} &= C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} \\ &\quad + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y}\Bigg[H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k}\left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta x}\right) \\ &\qquad - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k} - D_{b,i+\frac{1}{2},j-\frac{1}{2},k}\left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k}}{\Delta y} - \frac{E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}}{\Delta x}\right)\Bigg] \\ &\quad - \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z}\left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right). \end{aligned} \tag{3.7}$$

Collecting the unknown $E_x^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} &- \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}\, E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} \\ &+ \left[1 - \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} + \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}\right] E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ &- \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}\, E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} \\ = {}& C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y}\left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y\,\Delta x}\left(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\right) \\ &+ \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y\,\Delta x}\left(E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}\right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z}\left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right). \end{aligned} \tag{3.8}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} + \beta_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - \gamma_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} = p_{x1} E^{n}_{x,i+\frac{1}{2},j,k} \\
+ \frac{1}{\Delta y}\left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\
- \frac{1}{\Delta z}\left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right) \\
+ \frac{q_{x1}}{\Delta x \Delta y}\left(E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}\right) \\
- \frac{r_{x1}}{\Delta x \Delta y}\left(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\right),
\end{aligned} \tag{3.9}$$

and the magnetic update can be written as

$$H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + \frac{r_{x1}}{\Delta y}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right) - \frac{r_{x1}}{\Delta x}\left(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\right). \tag{3.10}$$

where

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \tag{3.11}$$

$$\beta_{x1} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{3.12}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \tag{3.13}$$

$$p_{x1} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.14}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{3.15}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.16}$$

E_y **and** H_x By equation (1.2b),

$$\varepsilon_{i,j+\frac{1}{2},k} \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k} \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} + E^{n}_{y,i,j+\frac{1}{2},k}}{2} = [H_2], \tag{3.17}$$

where

$$[H_2] = \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}. \tag{3.18}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E^{n+\frac{1}{2}}_{y} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E^{n}_{y} + [H_2]. \tag{3.19}$$

Hence

$$E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} = C_{a,i,j+\frac{1}{2},k} E^{n}_{y,i,j+\frac{1}{2},k} + C_{b,i,j+\frac{1}{2},k} [H_2], \tag{3.20}$$

By equation (1.3a),

$$\begin{aligned}
H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \Bigg(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} \\
- \frac{E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta y} \Bigg),
\end{aligned} \tag{3.21}$$

Substituting the magnetic update into (3.20) and eliminating $H_x^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} &= C_{a,i,j+\frac{1}{2},k} E^{n}_{y,i,j+\frac{1}{2},k} \\
&\quad + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\Bigg[H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}\left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} - \frac{E^n_{z,i,j+1,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}}{\Delta y}\right) \\
&\qquad - H^n_{x,i,j+\frac{1}{2},k-\frac{1}{2}} - D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}\left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1}}{\Delta z} - \frac{E^n_{z,i,j+1,k-\frac{1}{2}} - E^n_{z,i,j,k-\frac{1}{2}}}{\Delta y}\right)\Bigg] \\
&\quad - \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x}\left(H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^n_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right).
\end{aligned}
\tag{3.22}
$$

Collecting the unknown $E_y^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
&- \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} \\
&+ \left[1 + \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2}\right] E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \\
&- \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1} \\
= {}& C_{a,i,j+\frac{1}{2},k} E^n_{y,i,j+\frac{1}{2},k} + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z\, \Delta y}\left(E^n_{z,i,j+1,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}\right) \\
&+ \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z\, \Delta y}\left(E^n_{z,i,j+1,k-\frac{1}{2}} - E^n_{z,i,j,k-\frac{1}{2}}\right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x}\left(H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^n_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right).
\end{aligned}
\tag{3.23}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1} + \beta_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - \gamma_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} = {}& p_{y1} E^n_{y,i,j+\frac{1}{2},k} \\
&+ \frac{1}{\Delta z}\left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right) \\
&- \frac{1}{\Delta x}\left(H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^n_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right) \\
&+ \frac{q_{y1}}{\Delta y \Delta z}\left(E^n_{z,i,j+1,k-\frac{1}{2}} - E^n_{z,i,j,k-\frac{1}{2}}\right) \\
&- \frac{r_{y1}}{\Delta y \Delta z}\left(E^n_{z,i,j+1,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}\right),
\end{aligned}
\tag{3.24}
$$

and the magnetic update can be written as

$$
H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + \frac{r_{y1}}{\Delta z}\left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\right) - \frac{r_{y1}}{\Delta y}\left(E^n_{z,i,j+1,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}\right). \tag{3.25}
$$

where

$$\alpha_{y1} = \frac{D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2}, \tag{3.26}$$

$$\beta_{y1} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y1} + \gamma_{y1}, \tag{3.27}$$

$$\gamma_{y1} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2}, \tag{3.28}$$

$$p_{y1} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{3.29}$$

$$q_{y1} = D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}, \tag{3.30}$$

$$r_{y1} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{3.31}$$

E_z and H_y By equation (1.2c),

$$\varepsilon_{i,j,k+\frac{1}{2}} \frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}}{\Delta t/2} + \sigma_{i,j,k+\frac{1}{2}} \frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} + E^n_{z,i,j,k+\frac{1}{2}}}{2} = [H_3], \tag{3.32}$$

where

$$[H_3] = \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}. \tag{3.33}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_z^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_z^n + [H_3]. \tag{3.34}$$

Hence

$$E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} = C_{a,i,j,k+\frac{1}{2}}\, E^n_{z,i,j,k+\frac{1}{2}} + C_{b,i,j,k+\frac{1}{2}}\, [H_3], \tag{3.35}$$

By equation (1.3b),

$$\begin{aligned} H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \Bigg(& \frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} \\ & - \frac{E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta z} \Bigg), \end{aligned} \tag{3.36}$$

Substituting the magnetic update into (3.35) and eliminating $H_y^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} = {} & C_{a,i,j,k+\frac{1}{2}}\, E^n_{z,i,j,k+\frac{1}{2}} \\ & + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \Bigg[H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta z} \right) \\ & \qquad - H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}} - D_{b,i-\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^n_{x,i-\frac{1}{2},j,k+1} - E^n_{x,i-\frac{1}{2},j,k}}{\Delta z} \right) \Bigg] \\ & - \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}} \right). \end{aligned} \tag{3.37}$$

Collecting the unknown $E_z^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
&-\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}\,E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} \\
&+\left[1+\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}+\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}\right]E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \\
&-\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}\,E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}} \\
=\;& C_{a,i,j,k+\frac{1}{2}}\,E^n_{z,i,j,k+\frac{1}{2}}+\frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x}\left(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}}-H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right) \\
&-\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x\,\Delta z}\left(E^n_{x,i+\frac{1}{2},j,k+1}-E^n_{x,i+\frac{1}{2},j,k}\right) \\
&+\frac{C_{b,i,j,k+\frac{1}{2}}\,D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x\,\Delta z}\left(E^n_{x,i-\frac{1}{2},j,k+1}-E^n_{x,i-\frac{1}{2},j,k}\right) \\
&-\frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y}\left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}}-H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\right).
\end{aligned} \tag{3.38}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{z1}E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}}+\beta_{z1}E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}-\gamma_{z1}E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} = {}& p_{z1}E^n_{z,i,j,k+\frac{1}{2}} \\
&+\frac{1}{\Delta x}\left(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}}-H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right) \\
&-\frac{1}{\Delta y}\left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}}-H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\right) \\
&+\frac{q_{z1}}{\Delta z\Delta x}\left(E^n_{x,i-\frac{1}{2},j,k+1}-E^n_{x,i-\frac{1}{2},j,k}\right) \\
&-\frac{r_{z1}}{\Delta z\Delta x}\left(E^n_{x,i+\frac{1}{2},j,k+1}-E^n_{x,i+\frac{1}{2},j,k}\right),
\end{aligned} \tag{3.39}
$$

and the magnetic update can be written as

$$
H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}}+\frac{r_{z1}}{\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}}-E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right)-\frac{r_{z1}}{\Delta z}\left(E^n_{x,i+\frac{1}{2},j,k+1}-E^n_{x,i+\frac{1}{2},j,k}\right). \tag{3.40}
$$

where

$$
\alpha_{z1} = \frac{D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.41}
$$

$$
\beta_{z1} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}}+\alpha_{z1}+\gamma_{z1}, \tag{3.42}
$$

$$
\gamma_{z1} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.43}
$$

$$
p_{z1} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{3.44}
$$

$$
q_{z1} = D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}, \tag{3.45}
$$

$$
r_{z1} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{3.46}
$$